

networks. With SDN, managers may modify the network using the central controller, which can be done more quickly and with fewer chances of error.

SDN also gives the network more adaptability and programmability. Administrators can utilise software to develop unique policies and behaviours that are tailored to the requirements of their company because the central controller is in charge of defining the behaviour of the network. This can facilitate the support of new apps and services while also enhancing network performance and efficiency.

SDN, in its entirety, signifies a change in how networks are planned and operated, shifting the attention away from individual devices and towards the network. It provides many advantages, such as increased productivity, adaptability, and programmability, which can assist organisations in better meeting the changing needs of their users and company.

Control plane and forwarding

plane: The control plane and the forwarding plane (data plane), which are two distinct operational planes, collaborate to manage and route traffic via a network in an SDN architecture. The functionality of the network is managed and controlled by the control plane. It accomplishes this through communication with the central controller, the key element of the SDN architecture. The central controller is a piece of software that controls the network's policies and behaviours. It runs on a server or other specific device.

On the other side, the forwarding plane is in charge of actually moving traffic through the network. It accomplishes this by following the guidelines and directives established by the control plane. The forwarding plane is made up of networking components like switches and routers, which act as

straightforward forwarding elements and execute commands from the central controller. The control plane and the forwarding plane in an SDN architecture are decoupled, which means they work separately from one another. This can help to increase the effectiveness and performance of the network by allowing the control plane to concentrate on controlling the network and the forwarding plane to concentrate on forwarding traffic. To govern and route traffic via the network, the control plane and the forwarding plane are two essential parts of SDN architecture.

The forwarding plane oversees forwarding traffic based on the network's policies and behaviours, whereas the control plane is in charge of setting those policies and behaviours. The SDN architecture is detailed at <https://doi.org/10.1109/INDISCON54605.2022.9862876>.

Network functions virtualisation:

Network functions virtualisation (NFV) allows network functions such as firewalls, load balancers, and VPNs to be implemented in software and run on standard hardware instead of specialised equipment. This technology is often used in conjunction with software defined networking (SDN). Virtualised network functions or VNFs are created through software implementation and operate on conventional hardware servers. Virtual switches route traffic to and from the VNFs, and the management plane oversees and controls the network and VNFs, including monitoring performance, troubleshooting, and data collection.

NFV provides several benefits, including faster and easier installation and scalability of network functions, as well as greater flexibility and adaptability. VNFs are easily programmable and portable, which can facilitate changes to the network based on evolving needs.

Tacker: VNF life cycle management in OpenStack

Tacker is an open source project that is part of the OpenStack cloud computing platform. Its purpose is to simplify the deployment and maintenance of network functions virtualisation (NFV) in OpenStack setups. With NFV technology, network services such as firewalls, load balancers, and VPNs can be implemented in software and operated on common hardware rather than on specialised networking equipment. Tacker is a suite of tools and APIs that enable administrators to create and manage VNFs and their connections, streamlining the deployment and maintenance of NFV services in OpenStack systems.

Tacker is designed as a service that runs on the OpenStack cloud and is accessible through a web-based dashboard or command line interface. Resources needed to enable VNFs, such as virtual machines and switches, are automatically provisioned, and administrators can define VNFs and their connections using a declarative paradigm. Overall, Tacker is a valuable tool for organisations adopting OpenStack that want to leverage NFV, providing a range of tools and APIs that facilitate the deployment and administration of VNFs in OpenStack environments, thereby enhancing the network's flexibility and programmability.

The complete Tacker architecture is available at <https://docs.openstack.org/tacker/wallaby/user/architecture.html>.

Deploying a VNF in OpenStack

By deploying VNFs in OpenStack, organisations can enhance network agility, reduce hardware costs, and rapidly scale and adapt their network services.

VNF descriptor: A virtual network function (VNF) descriptor is a document or file that outlines the characteristics and behaviour of a virtualised network function.